Q1.		A healthcare start-up is designing an assistive mobile app for visually and motor-impaired users to help them book medical appointments, store prescriptions, and get medication reminders. The company aims to ensure accessibility through a Human-Centered Design (HCD) approach. **[Total Marks 20]**
	1.	List and briefly describe in one sentence three (03) user research methods suitable for understanding accessibility needs in this scenario. (3 marks)
	1. Conduct one-on-one interviews with visually and motor-impaired users to understand their needs, challenges, and expectations when using healthcare applications. 2. Observe users in their real environment while they manage healthcare tasks to identify accessibility barriers and their daily interaction patterns. 3. Allow users to interact with accessible prototypes to identify usability issues and gather feedback for improving the application's accessibility.	
		2. Explain why inclusive UX research is essential during the early design phase of accessibility-focused applications. (2 marks)
	Inclusive UX research helps designers understand the needs of users with different abilities from the beginning of the project. This ensures that accessibility issues are identified early, leading to a more usable, inclusive, and cost-effective application while reducing the need for major redesigns later.	
	3.	Explain why human centered design is crucial for accessibility refinement. (3 marks)
	Human-Centered Design focuses on the needs, abilities, and feedback of users throughout the design process. By involving users with disabilities in testing and evaluation, designers can identify accessibility issues early and continuously improve the application to make it more usable, effective, and inclusive.	

	4.	The app includes a voice command feature for scheduling appointments. Identify two (02) usability issues visually impaired users might face and propose one solution for each. (4 marks)

Usability Issue

Voice recognition may fail to understand different accents or unclear speech.

Users may not receive confirmation that the appointment was successfully scheduled.

Solution

Provide an option to repeat the command or allow manual input using accessible controls.

Give clear voice feedback and vibration/audio confirmation after completing the action.

	5.	Outline a usability-testing plan specifying the participant profile, two test tasks, and two performance metrics to assess accessibility and satisfaction. (4 marks)

User Profile	Test Task	Performance Metrics
Visually impaired users who use screen readers	Schedule a medical appointment using voice commands	Task completion rate, Time taken to complete the task

		Users with motor impairments	Set a medication reminder	Number of errors, User satisfaction rating
	6.	Mention two (02) ethical considerations related to testing with users with disabilities. (4 marks) 1. Participants should clearly understand the purpose of the study and voluntarily agree to participate before testing begins.		

		2. Protect participants' personal and medical information and ensure that all collected data remains confidential.
Q2		Use Figure 01 and Figure 02 to answer the following questions. Figures are annexed on the last page of the paper [**Total Marks 20]**
	1.	Identify strengths or effective UX elements in both designs that enhance clarity and user flow. (6 marks)

Figure 01 – Account Setup (Password Reset)

- Clear page title (**"Reset Your Password"**) helps users understand the purpose immediately.
- Input fields are labeled (Employee ID and Date of Birth), making the form easy to follow.
- **Continue** and **Cancel** buttons provide clear navigation options.

Figure 02 – eBay Product Details Page

- Large product image with multiple thumbnails allows users to view the product from different angles.
- Product information (price, condition, seller, size, and color) is organized into separate sections.
- Drop-down menus for selecting size and color simplify the purchasing process.

2.	Highlight weaknesses that could reduce usability or accessibility for new users. (6 marks)

Figure 01

- The year drop-down contains a very long list (2060–2078), making date selection slow and confusing.
- Buttons are small and have low visual emphasis, reducing accessibility for users with motor impairments.
- The form lacks clear error messages or guidance for incorrect input.

Figure 02

- Too much information is displayed at once, creating visual clutter for new users.
- Important actions such as selecting size and color may not stand out enough.
- Small text and crowded layout may be difficult for users with visual impairments.

	3.	Recommend three specific design improvements to enhance layout organization. (8 marks)

1. **Simplify the Form layout**

- Group related fields together and use a modern date picker instead of a long year drop-down to reduce user effort.

2. Improve visual hierarchy

- Increase the size of headings, buttons, and important information (such as price and action buttons) using better spacing and contrast.

3. Reduce clutter and improve spacing

- Add more white space between sections and organize information into clearly separated panels so users can scan the page more easily.

Q3	You are the UX designer for *"FoodEase"*, a smart meal-planning and food delivery application designed to help busy professionals organize weekly meal plans, order groceries, and track nutrition goals. Users have recently complained that the home dashboard feels crowded and they find it hard to identify key functions like "Plan Meal," "Order Now," and "Nutritional Summary." **[Total Marks 20]**	
	1.	How does cognitive overload affect user errors? Name common cognitive errors and mitigation techniques that can be used by designers(6 marks)

	1.	
	2.	
	2.	Define the term Brand guideline and explain how typography, color, and alignment could make *FoodEase*'s navigation more intuitive. (4 marks)

	3.	Identify two usability issues from the feedback and describe their effect on efficiency and task performance. (4 marks)
	4.	Using Nielsen's Usability Heuristics, propose two design improvements, explaining how each directly resolves the identified issues. (6 marks)

Q4	A design team is developing a wellness and productivity app called *"FocusPlus"* that helps users schedule tasks, monitor stress, and balance work-life activities. During early testing, users reported that the interface feels visually crowded, notifications are overwhelming, and color choices are inconsistent — causing distraction and confusion. [**Total Marks 20]**	
	1.	Subconscious processes refer to mental activities that occur automatically without deliberate awareness. These processes strongly influence perception, decision-making, and behavior during interaction with digital systems. Explain any of the related theory (Emotional Design Theory, Dual Process Theory, or cognitive biases) using the examples with *FocusPlus*. (6 Marks)

2.	Subconscious and mental processes that different users cannot always explain through their behavior, whereas researchers use different research methods on that. Provide two research methods to identify them and describe how to minimize them. (6 marks).

3.	Discuss the difference between the Gulf of Execution and the Gulf of Evaluation in this scenario. Suggest one design improvement for each to bridge these gaps. (4 marks)

	4.	Explain one attention management technique that can reduce distraction from constant notifications in *FocusPlus*. (4 marks)
Q5	With a scenario from your choice when required, answer the following questions **[Total Marks 20]**	
	1.	How to conduct brainstorming techniques efficiently and effectively. (4 Marks)

2.	What are the desirable aspects you need to consider while developing an Interface with current industry trends? Explain using an example situation? (8 Marks)

3.	Apply Shneiderman's Eight Golden Rules or Nielsen's 10 Principles to propose three corresponding design solutions that improve system usability of learning management platforms. (8 Marks)

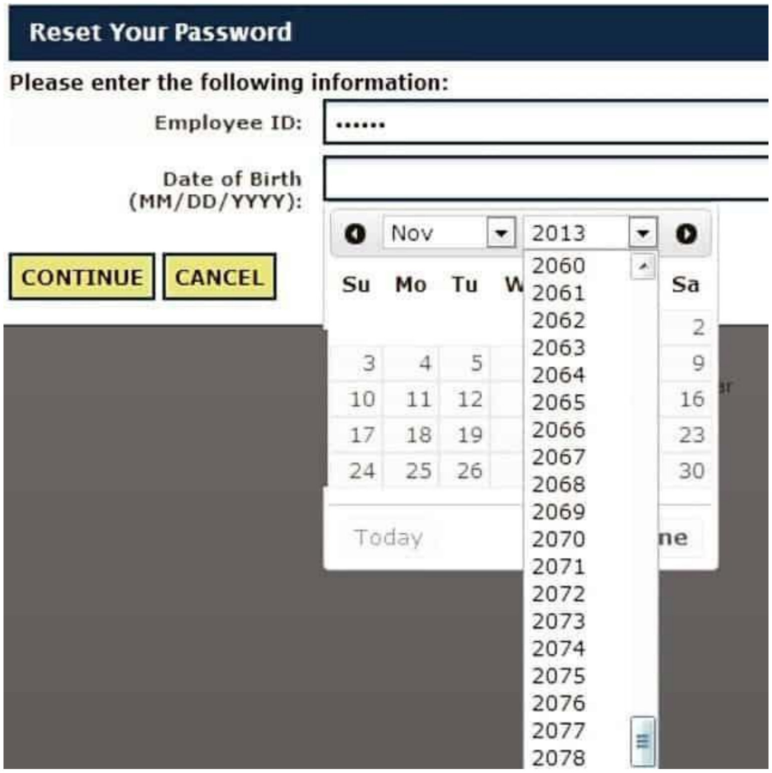

Figure 01: Account Setup for company employee profile

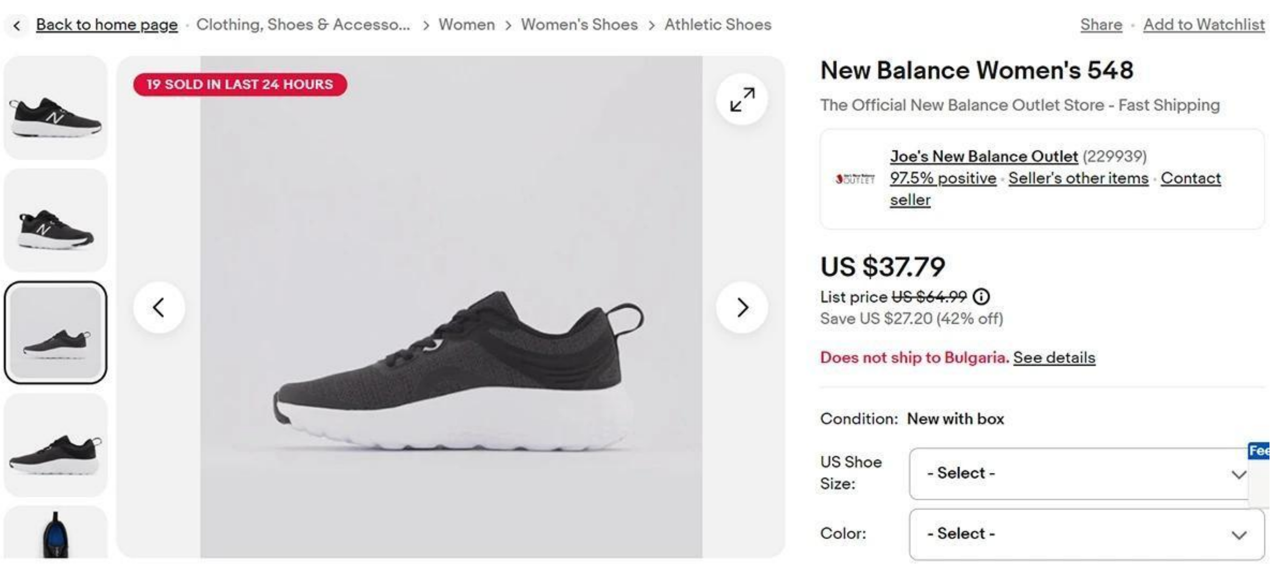

Figure 02: Ebay Product Details Page

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